

Our performance

Key figures at a glance

Economic performance indicators

2018 sales with social and environmental benefits

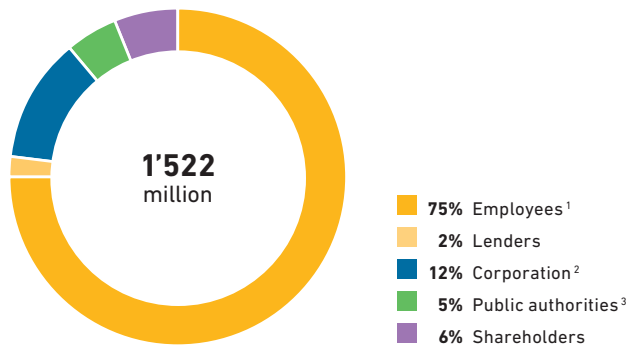
% of total GF sales

47%

Examples of identified benefits include but are not limited to: ensuring safe and hygienic potable water in buildings; safe and leak-free distribution of gas and chemicals to ensure fewer human and environmental accidents; light metals components, including components for electric or plug-in hybrid vehicles, helping to reduce the weight and CO₂ footprint of vehicles; solutions for our machining customers to produce their products in a more energy-efficient way.

Net value added 2018

% (100% = CHF 1'522 million)



1 Salaries and wages, employee benefits and social security

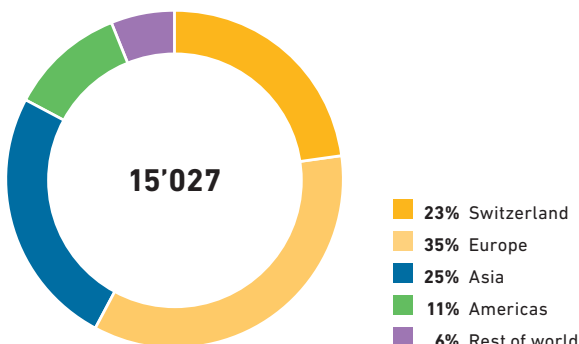
2 Retained earnings

3 Income taxes

Social performance indicators

Employees by region

% of headcount (100% = 15'027)



Accident rate GF employees

15.9 per 1 million working hours

Absence rate GF employees

4.1% of total work days

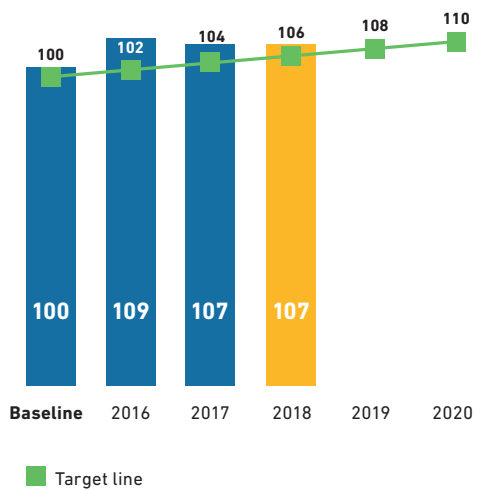
Total employee fluctuation

11.9%

Enviromental performance indicators

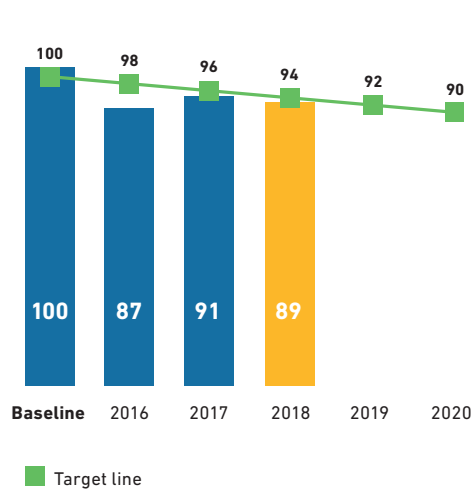
Energy efficiency index

(production volume/energy consumption) in %



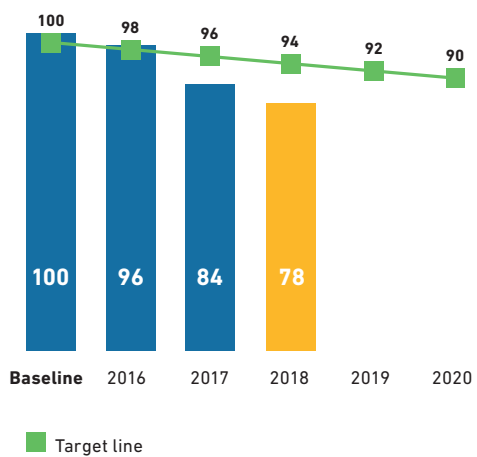
Emissions intensity index

(CO₂e emissions/production volume) in %



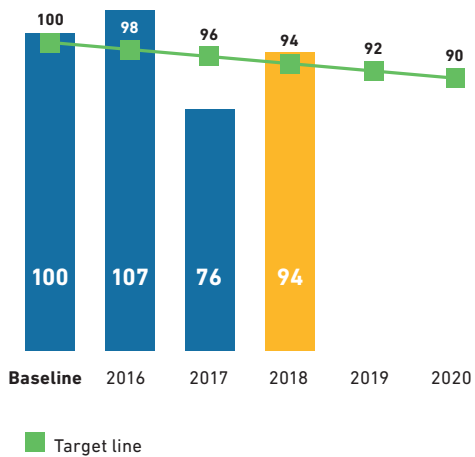
Non recycled waste intensity index

(non-recycled waste/production volume) in %



Water intensity index

(water consumption/production volume) in %



Retrospective

Highlights 2018

Eco-design in GF products and solutions

Staying true to our commitment to increasing the energy efficiency of machines for our customers, during 2018 GF Machining Solutions introduced a number of eco-design features across the entire product range. First, our specialists conducted an assessment of the biggest levers for energy optimization in machine design and usage.

The resulting measures included but were not limited to:

- a new spark generator on EDM machines, reducing energy consumption by over 30% at the same performance levels
- the reduction of up to 50% energy usage for part pro-

duction in laser machines due to the optimization of axis movements

- 50% reduction of high energy compressed air usage in milling models through new and patented developments
- rollout of the EconoWatt application for all new models to help save energy when the machine is idle



Energy-efficient cooling technology

In January 2018, a new cooling infrastructure was installed at GF Piping Systems in Schaffhausen. Used to chill injection molding machines, the new system, in combination with a few

other measures, is expected to save as much energy per year as would be needed to provide electricity and heating to 950 single family households (about 7 million kWh) on an annual basis.



LEED Silver certification for a GF Casting Solutions site

In June 2018, the new GF Linamar site in Mills River (USA) achieved Silver certification under the LEED 2009 New Construction and Major Renovations rating system (Leadership in Energy and Environmental Design). This is the first such case for a high-pressure die-casting facility in the world.

Intelligent use of resources

A new heat recovery system went into effect in early 2018 at the site of GF Casting Solutions in Leipzig (Germany). The installed air compressors are now able to process waste heat for warming of the production hall areas. As a result, approx. 422'000 kWh of heating power was provided during 2018 by applying this process instead of using fossil fuels.

Getting the safety culture anchored from an early age

As part of the broader Zero Risk campaign, the GF Casting Solutions site in Herzogenburg (Austria) conceptualized and conducted a pilot course for its trainees. In November, 23 youngsters participated in an interactive week-long program, which made them aware of the fundamentals of GF's safety standards and expectations of safe behavior at work and beyond. As part of the training, the apprentices conducted a project to eval-

uate safety-related measures on site and to provide fresh independent input for potential improvements.

The successful pilot training had an impact at the location, and it is aimed to be repeated on an annual basis, as well as rolled out at other locations.



ronmental, social and governance risks as well as opportunities in sourcing. The initial concept will be formalized during the first half of 2019.

Our employees

At the end of 2018, GF employed 15'027 people¹, which was 5% below the level of 2017. This was primarily due to the divestments² of the Singen and Mettmann (Germany) iron casting plants of GF Casting Solutions. At the same time, the number of female employees increased to 2'512 headcounts (or about 17% of the total workforce). Around 77% of GF employees participated in various professional development courses, on average 2.5 days per employee. Worldwide, 469 apprentices were employed and trained by GF in 2018.

Health and safety in the workplace

Workplace health and safety remained a key item on our agenda in 2018. During the reporting year, we observed a decrease in the number of severe cases, but unfortunately saw a rise in the overall number of accidents (425 compared to 407 in 2017). However, as the number of working hours performed in 2018 increased as well, the accident rate per one million working hours remained unchanged at 15.9 on a Corporation-wide level.

To facilitate a better overview of accidents that still happen at GF and to be able to quicker identify areas requiring safety improvement, we rolled out a Corporation-wide accident-reporting

platform. Additionally, during 2018, all sites of GF Piping Systems organized safety events focusing on firefighting training, hand and finger safety as well as other site-specific safety topics.



Safety event at GF Piping Systems in Sissach (Switzerland)

Besides raising awareness, all three Divisions conducted cross-site-safety audits at the vast majority of GF production plants. One of the benefits of this process is the exchange of best practice examples among the locations. The audit findings will serve as a basis for defining specific forward-looking actions to improve workplace safety.

The absence rate increased in 2018 to 4.1% compared to 3.96% in 2017. The majority of absence days were due to non-work-related illness, with only 6% having work-related causes.

¹ Headcount figure

² This was partially counter-balanced by the addition of new employees from the new facilities in Mills River (USA) and the acquisition of Precicast (Switzerland).



Ambient air cooler installation on the roof of the GF Piping Systems plant in Dautphetal (Germany).

Environmental efficiency

As a result of a solid order pipeline, production ran at full capacity in many of our factories in 2018, and sales grew organically by 7%. At the same time, this growth was accompanied by a relatively small increase in GF's net energy consumption (2.5%). Total waste was the same as in 2017, and water consumption for the year decreased by 5.1%.

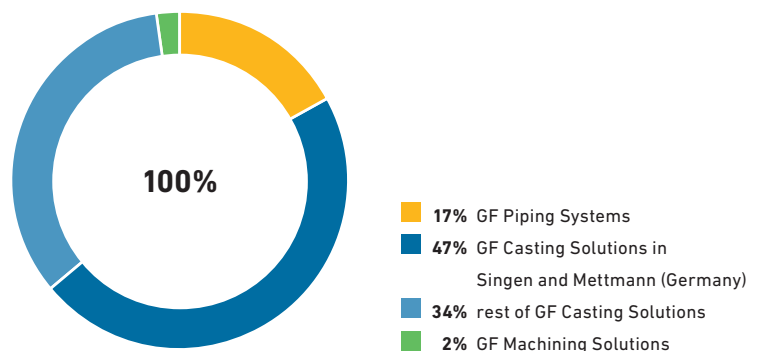
Due to the continuous implementation of smaller-to-medium-impact efficiency measures at our production plants worldwide, GF maintained its energy efficiency rate in 2018 at 107³, which resulted in a seamless achievement of the level expected for 2018. GF Piping Systems stood out among the three Divisions by systematically screening their production processes in order to find potential optimization measures, such as cooling with ambient air or investments in new more efficient machines.

Overall CO₂-equivalent (CO₂e) emissions dropped by 1.7%, thereby enabling GF to achieve its emission intensity target for 2018. This was possible due to a combination of more efficient energy consumption and the purchase of green energy certificates for selected locations of GF Casting Solutions and GF Machining Solutions. For the reporting year, the share of green

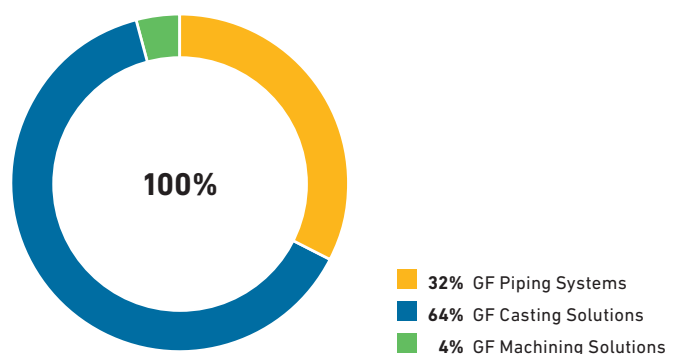
energy⁴ was 8.76% of the total energy consumption of GF, slightly below the 2017 level (10%).

In the coming years, GF's focus in the sphere of environmental management will be on further reducing its energy consumption and the resulting GHG emissions. Some of the areas that we are already reviewing are possible energy efficiency improvements for casting machines used by GF Casting Solutions⁵, as well as solutions for green electricity for GF Piping Systems and GF Machining Solutions.

Divisional share of net energy consumption in 2018 in %



Divisional share of net energy consumption after divestment of Singen and Mettmann in %



³ Calculated as a % of production volume over energy consumption

⁴ Including green electricity

⁵ Workshops with key direct suppliers of GF Casting Solutions are being organized for 2019

With the divestment of two iron-casting sites of GF Casting Solutions in Germany, the divisional share of the environmental footprint of GF will change. Due to the nature of its production processes, GF Casting Solutions will still account for the largest share of GF's energy consumption, while the share accounted for by GF Piping Systems and GF Machining Solutions will increase.

Management involvement

As a crucial measure to further integrate sustainability into the daily activities of GF, at the end of 2018, we defined specific targets in the 2019 personal goals of each member of the Executive Committee of the Corporation. The specific points match and support the strategic efforts of each Division and are being cascaded into the organization.

Outlook

As an outlook for the next year and preparing for the new business strategy cycle, we will define a long-term aspiration level (2030+) for sustainability at GF as well as specific goals for the years 2021-2025 on integrating the sustainability topic into the respective strategies of GF and its Divisions.



Andreas Müller, CFO of GF, opening the Sustainability Conference Europe in Zurich (Switzerland) in September 2018.

Economic performance indicators

CHF million	2018	2017	2016	2015	2014
Order intake	4'521	4'274	3'749	3'662	3'836
Orders on hand at year-end	623	773	614	612	634
Income statement					
Sales	4'572	4'150	3'744	3'640	3'795
EBITDA	529	491	443	422	399
Operating result (EBIT)	382	352	311	296	274
Net profit/loss after minorities	281	252	216	188	184
Cash flow					
Cash flow from operating activities	397	410	400	328	248
Cash flow from investing activities	-404	-280	-265	-140	-158
Free cash flow before acquisitions/divestitures	147	204	231	190	110
Free cash flow	-7	130	135	188	90
Balance sheet					
Assets	3'444	3'610	3'202	3'083	2'989
Liabilities	2'016	2'241	2'002	1'953	1'885
Equity	1'428	1'369	1'200	1'130	1'104
Net working capital	926	899	838	819	864
Invested capital (IC)	1'494	1'466	1'333	1'279	1'354
Net debt	238	183	214	238	354
Key figures					
Return on equity (ROE) %	19.9	20.1	19.3	17.7	18.7
Return on invested capital (ROIC) %	22.4	20.3	19.3	18.9	17.9
Return on sales (EBIT margin) %	8.4	8.5	8.3	8.1	7.2
Asset turnover	3.1	3.0	2.9	2.8	2.9
Cash flow from operating activities in % of sales	8.7	9.9	10.7	9.0	6.5
Employees					
Employees at year-end	15'027	15'835	14'808	14'424	14'140
Europe	8'721	9'658	8'845	8'783	8'676
Asia	3'725	3'807	3'713	3'502	3'455
Americas	1'740	1'503	1'348	1'262	1'259
Rest of world	841	867	902	877	750

Environmental performance indicators¹

	Unit	2018 ²	2017 pro rata ²	2017	2016	2015	2014
Energy							
Gross energy consumption	1'000 GJ	6'823	6'670	6'941	6'605	6'392	6'317
Electricity	1'000 GJ	3'249	3'239	3'296	3'168	2'878	2'835
Natural gas, biogas, fuel oil	1'000 GJ	1'363	1'218	1'247	1'205	1'262	1'186
Coke, lignite	1'000 GJ	2'092	2'084	2'265	2'121	2'156	2'192
Fuel use (for internal transportation)	1'000 GJ	97	106	110	92	80	72
Other energy sources	1'000 GJ	22	23	23	19	16	30
Energy sold	1'000 GJ	-82	-95	-103	-96	-105	0
Net energy consumption	1'000 GJ	6'742	6'575	6'838	6'509	6'287	6'317
Renewable energy (incl. green electricity)	%	9	10	10	11	11	11
Target line energy efficiency index ³ (production volume/energy consumption ⁴)		106	104		102	100	
Energy efficiency index ² (status at year-end)		107	107		109	100	
GHG emissions (in CO₂e)⁵							
Total CO ₂ e emissions	1'000 tonnes	630	641	668	610	580	580
Scope 1 (Direct emissions: energy consumption)	1'000 tonnes	312	304	325	307	313	313
Scope 2 (Indirect emissions: electricity and district heating)	1'000 tonnes	312	330	337	299	264	264
Scope 3 (Indirect emissions: business travel)	1'000 tonnes	6	7	7	4	3	3
Target line GHG emission intensity index ³ (CO ₂ e emissions/production volume ⁴)		94	96		98	100	
GHG emission intensity index ² (status at year-end)		89	91		87	100	
Air emissions							
Nitrogen oxides (NO _x)	1'000 tonnes	0.05	0.04	0.04	0.03	0.03	0.03
Sulphur oxides (SO _x)	1'000 tonnes	0.01	0.01	0.01	0.01	0.01	0.01
Water and wastewater							
Total water consumption	1'000 m ³	2'759	2'907	2'966	2'808	2'875	2'712
City water from public supply	1'000 m ³	697	657	662	633	653	596
Ground and rain water	1'000 m ³	2'062	2'250	2'304	2'175	2'222	2'116
Wastewater volume	1'000 m ³	1'961	2'077	2'098	1'977	1'961	878
Wastewater to sewage systems	1'000 m ³	814	926	944	897	890	830
Wastewater returned to nature, unpolluted	1'000 m ³	1'147	1'151	1'153	1'081	1'071	47
Target line water intensity index ³ (water consumption/production volume ⁴)		94	96		98	100	
Water intensity index (status at year-end)		94	76		107	100	

	Unit	2018 ²	2017 pro rata ²	2017	2016	2015	2014
Waste and recycling							
Total waste	1'000 tonnes	306	306	326	314	309	315
Normal waste, recycling	1'000 tonnes	245	246	263	249	233	250
Normal waste, landfill or incineration	1'000 tonnes	43	43	44	45	54	43
Hazardous waste	1'000 tonnes	18	17	18	20	22	21
Hazardous waste, recycling	1'000 tonnes	16	15	16	14	15	15
Hazardous waste, storage or incineration	1'000 tonnes	2	2	2	6	6	6
Target line non-recycled waste intensity index ² (non-recycled waste/production volume ⁴)		94	96		98	100	
Non-recycled waste intensity index ² (status at year-end)		78	84		96	100	
Business trips							
Air travel	1'000 pkm	31'379	36'850	36'872	31'863	14'305	22'342
Monetary values							
Expenditure for environmental protection	CHF million	18	16	18	15	16	13
Energy costs	CHF million	130	124	128	103	118	131
Water and wastewater costs	CHF million	4	3	3	3	3	3
Waste disposal costs and recycling credits	CHF million	9	7	8	8	7	9

1 The environmental performance indicators include all GF production companies.

2 Due to the divestment (as of 1 December 2018) of the Singen and Mettmann plants of GF Casting Solutions, the 2018 environmental performance figures here are presented according to the effective consumption at these sites while part of the GF Corporation (11 months of 2018). To ensure data consistency and comparability over time, the pro-rata 2017 data column shows how comparable consumption would have been in 2017. In the calculation of the target-relevant environmental indicators for all the years in this overview, we applied the respective 11/12th ratio for the two divested sites.

3 The target line is calculated linearly based on the standardized baseline consumption (=100) and according to the defined target.

4 The production volume is defined based on the specifics of the businesses of the individual Divisions: as "produced tonnes" for GF Piping Systems, as "gross value added" for GF Casting Solutions and as "hours worked" for GF Machining Solutions.

5 CO₂-equivalents is a unit used to transform other greenhouse gases into CO₂e (CO₂=1) according to their global warming potential. The applied emission factors are based on the GHG Protocol 2016.

Social performance indicators¹

	Unit	2018 ²	2017	2016	2015	2014
Employees						
Number of employees	Headcount	15'027	15'835	14'808	14'424	14'140
	FTE ³	14'413	15'163	14'105	13'783	13'518
Female employees	Headcount	2'512	2'347	2'245	2'162	2'045
	FTE	2'397	2'213	2'055	1'805	
Student interns	Headcount	174	145	182	154	213
Apprentices	Headcount	469	525	540	509	506
Women in management positions ⁴	FTE	88	90	95	87	84
	% of total management positions	15	14	17	15	15
Women on the Executive Committee	Number	0	0	0	0	0
Women on the Board of Directors	Number	2	2	2	2	2
	%	22	22	22	22	22
Total departures	Headcount	1'796	1'603	1'512	1'127	1'393
Total employee fluctuation	%	11.9	11.3	11.6	8.9	11.0